TEAM-9 BANKING MANAGEMENT SYSTEM

UML DIAGRAMS & SYSTEM ARCHITECTURE SPECIFICATION

Coursework Documentation • StarUML / Rational Rose / PlantUML Standard

1. Diagram Deliverables & Source Files

Diagram Name	Dedicated File (.puml)	Compatible Tools
UML Class Diagram	*diagrams/ClassDiagram.puml*	StarUML, Rational Rose, PlantText
UML Activity Diagram	*diagrams/ActivityDiagram.puml*	StarUML, Rational Rose, PlantText
3-Tier System Architecture	*diagrams/SystemArchitecture.puml*	Architecture Report & Presentation
UML Use Case Diagram	*diagrams/UseCaseDiagram.puml*	StarUML / Rational Rose Use-Case View
UML Sequence Diagram	*diagrams/SequenceDiagram.puml*	StarUML / Rational Rose Sequence View
Combined All Diagrams	*diagrams/ALL_UML_DIAGRAMS.puml*	Instant 1-Click Paste for StarUML

2. UML Class Diagram - Domain Entities & Attributes

Class Name	Attributes & Data Types	Operations & Methods
User	+ id: String + fullName: String + email: String + phoneNumber: String - passwordHash: String + role: Role	+ register(data): User + login(credentials): SessionToken + resetPassword(newPass): Boolean
Account	+ id: String + accountNumber: String + upiId: String - upiPinHash: String + failedPinCount: Integer + pinLockedUntil: DateTime + balance: Decimal + status: AccountStatus	+ checkBalance(amt): Boolean + verifyPin(inputPin): Boolean + lockPin(duration): Void + unlockPin(): Void + updateBalance(delta): Decimal
Transaction	+ id: String + referenceNumber: String + idempotencyKey: String + amount: Decimal + fee: Decimal + status: TransactionStatus + category: TransactionCategory + senderBalanceAfter: Decimal + receiverBalanceAfter: Decimal	+ executeAtomicTransfer(): Result + generateReferenceNumber(): String
Beneficiary	+ id: String + nickname: String + beneficiaryUpi: String + beneficiaryName: String + ownerId: String	+ validateRecipient(): Boolean + saveBeneficiary(): Beneficiary + deleteBeneficiary(): Boolean
Notification	+ id: String + title: String + message: String + type: String + isRead: Boolean	+ markAsRead(): Void + sendEmailAlert(): Void
AuditLog	+ id: String + action: String + ipAddress: String + userAgent: String + details: Json	+ logEvent(action, details): Void

3. UML Activity Diagram - Money Transfer Flow (Question 42-45)

Activity Workflow Step	System Action & Security Rule
Step 1: Recipient Input	Customer inputs target UPI ID (e.g. priya@bank) or picks from saved beneficiaries.
Step 2: Recipient Resolution	Asynchronous DB query checks if account exists and is ACTIVE. Prevents self-transfer.
Step 3: Name Verification	Question 42 compliance: Displays verified recipient name (Priya Sharma) in green.
Step 4: Limits Validation	Validates: Amount >= INR 1.00, Amount <= INR 1,00,000, Daily utilized <= INR 2,00,000.
Step 5: Balance Check	Verifies available balance >= transfer amount. Rejects if funds insufficient.
Step 6: PIN Security Check	Customer enters 6-digit PIN. Salted bcrypt validation. 3 failed attempts lock account for 15 mins.
Step 7: ACID Row-Level Lock	Executes SELECT FOR UPDATE on Account rows in deterministic order to prevent race conditions.
Step 8: Atomic Balance Updates	Deducts amount from sender and credits receiver in single commit.
Step 9: Ledger & Audit Logs	Generates TXN-YYYYMMDD-XXXXXX, inserts Transaction record and security AuditLog.
Step 10: Alerts & Receipt	Emits in-app notifications and email alerts. Displays success receipt with reference ID.

4. Enterprise 3-Tier System Architecture

Architectural Tier	Technology Stack & Responsibilities
Tier 1: Presentation Layer	Next.js 16 (App Router), React 19, Tailwind CSS v4, Accessible Dialogs, Responsive Dashboard, Admin Console (/admin), PDF/CSV Statement Exporters.
Tier 2: Business Logic Layer	Next.js Server Actions, Zod input validation, JOSE JWT session tokens (HTTP-only secure cookies), bcryptjs password & 6-digit PIN hashing, Core Banking atomic engine, Notification dispatcher.
Tier 3: Data & Persistence Layer	PostgreSQL Relational Database, Prisma ORM with automated migrations, Row-Level Locking (SELECT ... FOR UPDATE) for zero double-spending, Immutable AuditLog table.